

THAMIRES ANDRADE LIMA, Ph.D.

R&D Senior Scientist Candidate | Spectroscopy | Materials Characterization | Analytical Problem Solving
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TARGETED PROFILE

Experimental materials scientist with a Ph.D. in Physics and 18+ years of analytical laboratory experience across vibrational spectroscopy, X-ray/scattering methods, thermal/mechanical analysis, rheology, polymers, soft matter, complex fluids, fibers, gels, foams, biomaterials, and multicomponent materials. Strong fit for DuPont ADS spectroscopy role through deep Raman/FTIR experience, structure-property analysis, method development, data interpretation, technical reporting, SOP/user training, and cross-functional collaborations with academic and industry partners. Experienced translating complex measurements into actionable conclusions for product development, processing, manufacturing, and customer-facing materials challenges.

DUPONT ADS FIT

Background

- Raman/FTIR expertise applied to ionic liquids, polymers, peptide/fibrillar systems, biomaterials and complex fluids
- X-ray diffraction/scattering, neutron scattering, DSC/TGA, DMA, microscopy, rheology and mechanical testing
- Industry-facing projects connecting formulation, processing, sprayability, injection/filling behavior and packaging-material performance
- SOPs, user training, traceable experimental records, technical reports, data analysis and multidisciplinary communication

CORE TECHNICAL SKILLS

Spectroscopy & Imaging

Raman spectroscopy; low-frequency Raman; FTIR; vibrational spectroscopy; microscopy; chemical/structural interpretation.

Rheology & Processing

Shear and extensional rheology; non-Newtonian fluids; filament breakup; sprayability; injection/filling behavior; fiber spinning; nanofiber alignment; formulation-processing-property relationships.

Materials & Polymer Characterization

X-ray diffraction/scattering; neutron scattering; DSC/TGA; DMA; mechanical testing; thermal history; phase transitions; crystallization; glass formation; polymer/additive compatibility.

Data, Methods & Communication

Experimental design; data analysis/visualization; method development; SOPs; user training; technical reports; reproducible workflows; cross-functional presentations.

PROFESSIONAL EXPERIENCE

Research Assistant Professor | Drexel University, Department of Chemical and Biological Engineering | 2021-Aug 2026 | Philadelphia, PA

- Led applied materials research spanning polymers, complex fluids, soft matter, fibers, gels, foams, additive manufacturing resins, biomaterials, and structure-property relationships relevant to industrial formulation, packaging, and process-performance challenges.
- Applied Raman/FTIR spectroscopy, X-ray diffraction/scattering, rheology, DMA, DSC/TGA, mechanical testing, microscopy, and complementary analytical methods to connect chemistry, structure, processing, and performance.
- Developed antimicrobial active-packaging concepts using essential oils embedded in polymer networks, emphasizing polymer-additive compatibility, formulation-processing-property relationships, and packaging-material performance.
- Collaborated with companies on confidential projects involving shear and extensional rheology of complex products, improving understanding of injection/filling behavior, sprayability, formulation performance, and process compatibility.
- Designed process-relevant characterization workflows for non-Newtonian and polymeric systems where extensional deformation, filament breakup, atomization, and stretching behavior are more predictive than shear viscosity alone.
- Generated technical reports, experimental plans, SOPs, user-training materials, and reproducible workflows; communicated results to academic, industrial, and cross-functional stakeholders.

Postdoctoral / Research Scientist | Drexel University and Collaborating Institutions | 2019-2021 | Philadelphia, PA

- Advanced experimental programs on nanofibers, soft materials, peptide/fibrillar systems, gels, hydrogel foams, thermosets, and complex fluids using spectroscopy, rheology, thermal analysis, and mechanical testing.
- Investigated processing-structure-property relationships in polymeric and fibrous materials, including alignment, diameter control, annealing, thermal kinetics, and performance implications.
- Worked with interdisciplinary teams and external collaborators to translate laboratory measurements into practical conclusions for materials performance, formulation design, and application-oriented development.

Postdoctoral Scientist | University of Sao Paulo, UIUC, NIST, and Collaborative Programs | 2014-2018 | Brazil / USA

- Built deep expertise in physical chemistry and materials characterization, including Raman spectroscopy, X-ray diffraction/scattering, neutron scattering, low-temperature/high-pressure methods, and phase-transition analysis.
- Studied ionic liquids, glass-forming materials, confined systems, molecular crystals, and amino-acid materials, linking molecular structure and dynamics to crystallization, glass formation, relaxation behavior, and macroscopic properties.

- Trained instrument users and supported a multiuser experimental facility, strengthening experience in SOPs, safety, troubleshooting, method reproducibility, and technical communication.

SELECTED APPLIED RESEARCH CONTRIBUTIONS

- Vibrational spectroscopy and materials structure: Used Raman/low-frequency Raman and FTIR to study molecular structure, phase transitions, crystallization, glass formation, peptide/fibrillar assemblies, and polymeric/complex-fluid systems.
- Filament Extension Atomizers: Designed and executed rheology experiments and interpreted high-viscosity, non-Newtonian sprayability data to relate shear/extensional behavior to filament breakup and industrial spray-drying performance.
- Unexpected solidlike fracture in liquids: Connected rheology, stress analysis, and visual failure morphology to identify critical fracture behavior in high-viscosity simple liquids.
- Polymer/fiber platforms: Supported fiber spinning, post-drawing, nanofiber yarn fabrication, alignment, annealing, thermal/mechanical testing, and performance-driven optimization.

EDUCATION

Ph.D., Physics | Federal University of ABC (UFABC) | Brazil

M.S., Physics | Federal University of ABC (UFABC) | Brazil

B.Sc., Physics | Federal University of Sergipe (UFS) | Brazil

SELECTED PUBLICATIONS

- Unexpected Solidlike Fracture in Simple Liquids. *Physical Review Letters* 136, 124002 (2026).
- Filament Extension Atomizers: Integration Potential into Spray Dryers for High-Viscosity and Non-Newtonian Fluids. *Physics of Fluids* 38, 023331 (2026).
- Determining the Nanostructure and Main Axis of Gly-His-Gly Fibrils Using the Amide I' Bands in FTIR, VCD, and Raman Spectra. *Spectrochimica Acta A* 306, 123584 (2024).
- High Performance Thermosets for Additive Manufacturing. *Innovation and Emerging Technologies* 10, 2330003 (2023).
- Vibrational Spectroscopy and MD Simulation of Choline Oxyanion Salts. *Journal of Molecular Liquids* 340, 117100 (2021).
- An inelastic neutron scattering, Raman, far-infrared, and molecular dynamics study of the intermolecular dynamics of two ionic liquids. *Physical Chemistry Chemical Physics* 22, 9074–9085 (2020).
- Low-Frequency Raman Spectra of a Glass-Forming Ionic Liquid at Low Temperature and High Pressure. *Journal of Chemical Physics* 150, 164502 (2019).
- Glass Transition and Melting Lines of an Ionic Liquid. *Journal of Chemical Physics* 148, 171101 (2018); Cover and Editor's Pick.
- Phase Transitions of [C2C1im][NTf2] under high pressure. A Synchrotron X-ray Diffraction and Raman Microscopy Study. *Crystal Growth & Design* 17, 5384 (2017).

ADDITIONAL

Applied Raman spectroscopy across 19 peer-reviewed publications, spanning phase transitions, glass formation, ionic liquids, molecular crystals, amino-acid systems, fibrillar materials, and structure–property characterization.

Publications: 35+ peer-reviewed articles

Presentations: 10+ oral talks and 20+ posters at conferences in Brazil, the United States, Argentina, Portugal, France, and Ukraine.

Experience developing interdisciplinary collaborations across Drexel, UPenn, Rowan, University of São Paulo, NIST, national laboratories, and industrial partners.

Mentoring: undergraduate, graduate, and postdoctoral training

Patent number: WO2024059060A1, Dyeing method for coloration of ultra-high molecular weight polyethylene fiber

Finalist of the ScientistA Award for Brazilian Women Scientists in the U.S. 2023.

Referee of the ScientistA Award for Brazilian Women Scientists in the U.S. 2024.

Reviewer for the *Frontiers in Physics*

Reviewer for the *Frontiers in Chemistry*

Reviewer for the *Applied Physics Letters*

Editor guest of the special issue “Preparation, Application and Performance Improvement of Hydrogel” of *materials* Journal.